Lab 2: Rigid-Body Transformations and Relative Trajectories in ROS

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***Abstract*— The MIT VNAV Lab 2 two-drone scenario is implemented in ROS Noetic. A transform broadcaster publishes the time-varying poses of two aerial vehicles at 50 Hz, while a second node queries TF and publishes their world-frame and relative trajectories for RViz. The observed paths are a circle for AV1, a parabola for AV2, and a planar ellipse for AV2 expressed in AV1. The report also derives the relative homogeneous transformation, identifies the supporting plane and ellipse semi-axes, and proves the requested quaternion operator identities.**

I. Introduction

The experiment studies how ROS TF represents rigid-body motion and composes reference frames. The prescribed world-frame positions are

$$
\begin{aligned}
{}^{w}\mathbf{o}_1(t) &= \begin{bmatrix}\cos t & \sin t & 0\end{bmatrix}^{\mathsf{T}}, \\
{}^{w}\mathbf{o}_2(t) &= \begin{bmatrix}\sin t & 0 & \cos 2t\end{bmatrix}^{\mathsf{T}}.
\end{aligned} \tag{1}
$$

AV1 has roll and pitch equal to zero and yaw equal to t, so its y axis is tangent to its circular path. AV2 translates without rotating relative to the world. The experiment covers TF publication and lookup, relative-trajectory analysis, and quaternion operator properties [1]. The same AV2 motion looks different in the world and AV1 frames: the world-frame path is an arc of a parabola, whereas the body-frame path is a closed ellipse. Keeping the two frames distinct is essential when interpreting the RViz trails.

II. ROS System and Experimental Method

A. Nodes, topics, and launch modes

In the static launch mode, the relevant nodes are `/av1broadcaster`, `/av2broadcaster`, `/plots_publisher_node`, and `/rviz`; `/rosout` is omitted. The two broadcasters publish `world→av1` and `world→av2` on `/tf_static`. The plot node subscribes to `/tf` and `/tf_static`, then publishes a `visualization_msgs/MarkerArray` on `/visuals`. RViz subscribes to the transform topics and `/visuals`; the latter carries the drone meshes and trajectory markers.

The same static system can be started without the launch file by running, in separate terminals,

```
roscore
rosrun tf2_ros static_transform_publisher
1 0 0 0 0 0 1 world av1
rosrun tf2_ros static_transform_publisher
0 0 1 0 0 0 1 world av2
rosrun two_drones_pkg plots_publisher_node
rosrun rviz rviz -d
$(rospack find two_drones_pkg)/config/
default.rviz
```

The launch argument `static` defaults to `false`. The `if` group therefore starts the two static broadcasters only when `static:=True`; the `unless` condition starts `/frames_publisher_node` only in dynamic mode. Omitting the argument consequently replaces both fixed broadcasters with the dynamic 50 Hz publisher.

The resulting dynamic ROS graph is shown in Table I.

TABLE I
ROS GRAPH IN DYNAMIC MODE (EXCLUDING /ROSOUT).

Node	Publishes	Subscribes
`frames`	`/tf`	–
`plots`	`/visuals`	`/tf`, `/tf_static`
`rviz`	–	`/visuals`, `/tf`, `/tf_static`

B. Transform publication and lookup

The implementation publishes

$$
\begin{aligned}
{}^{w}\mathbf{T}_1(t) &= \begin{bmatrix}\mathbf{R}_z(t) & {}^{w}\mathbf{o}_1(t) \\ \mathbf{0}^{\mathsf{T}} & 1\end{bmatrix}, \\
{}^{w}\mathbf{T}_2(t) &= \begin{bmatrix}\mathbf{I}_3 & {}^{w}\mathbf{o}_2(t) \\ \mathbf{0}^{\mathsf{T}} & 1\end{bmatrix}.
\end{aligned} \tag{2}
$$

The yaw quaternion is created by `setRPY(0,0,t)`; AV2 uses the identity quaternion. Both messages share the same timestamp and are broadcast every 0.02 s. The translation and rotation are placed in the same `TransformStamped` message for each vehicle. Using a common timestamp for the two transforms lets the downstream lookup describe both poses at a single instant. The 50 Hz publication rate describes how often TF is updated; the curves analyzed below come from the prescribed continuous-time equations, not from a fit to those updates.

For a trajectory requested as destination frame `dest_frame` expressed in reference frame `ref_frame`, the plot node calls

```
lookupTransform(ref_frame, dest_frame,
ros::Time(0)).
```

This TF convention returns the source pose in the target frame [2]. Three bounded trails are stored: AV1 in world, AV2 in world, and AV2 in AV1. The third is rendered as a dashed line.

III. Experimental Results

The package was compiled successfully in a ROS Noetic environment using `catkin build`. Figure 1 shows the static mode: RViz reports a valid world fixed frame, both

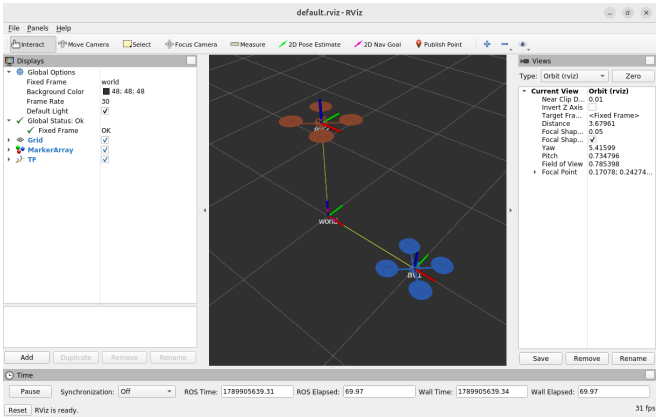


Fig. 1. RViz visualization in static mode with both meshes and TF frames.

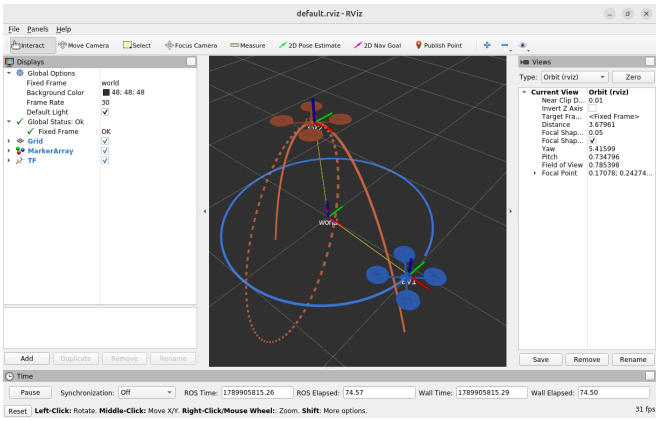


Fig. 2. Dynamic trajectories visualized in RViz with the fixed frame set to `world`.

vehicle meshes are visible, and TF axes connect `world`, `av1`, and `av2`.

In dynamic mode, Fig. 2 shows three trajectories. The blue solid curve is AV1's unit circle in the world x-y plane. The orange solid curve is AV2's parabola in the world x-z plane. The orange dashed curve is the AV2 trajectory expressed in AV1.

Figure 3 separates the two world-frame motions into the planes where they are defined. AV1 remains at $z_w = 0$ and has a unit-radius circular projection. AV2 remains at $y_w = 0$; its horizontal coordinate ranges from -1 to 1, while its height ranges from -1 to 1. The black markers indicate $t = 0$. These are analytic plots of Eq. (1), included to make the geometry in the RViz image easier to read.

Changing the RViz fixed frame to `av1` makes AV1 stationary while the world and AV2 move around it, as shown in Fig. 4. The relative dashed curve forms a closed curve on a slanted plane; its geometry is derived in Section IV.

Table II lists six saved samples obtained from `tf_echo av1 av2`: three from an initial capture and three from a second saved log. Their residuals in $z = 2y + 1$ are at most 0.001 when computed from the displayed three-decimal coordinates. The table describes the printed TF output rather than a higher-precision estimate of tracking error.

TABLE II

SAVED AV2 POSITIONS IN THE AV1 FRAME AND PLANE RESIDUALS. THE UPPER AND LOWER GROUPS ARE FROM SEPARATE TF CAPTURES.

x	y	z	$z-(2y+1)$
-1.149	-0.977	-0.955	-0.001
-1.479	-0.356	0.288	0.000
-0.682	-0.114	0.771	-0.001
-1.354	-0.147	0.705	-0.001
-0.666	-0.128	0.743	-0.001
-0.801	-0.959	-0.918	0.000

IV. Mathematical Analysis

A. AV2 world-frame parabola

From Eq. (1), $x = \sin t$, $y = 0$, and

$$z = \cos 2t = 1 - 2\sin^2 t = 1 - 2x^2. \tag{3}$$

Thus AV2 lies on the parabola $z = 1 - 2x^2$ in the world x-z plane. As t runs from 0 to 2π, $x = \sin t$ visits both positive and negative halves of this arc and retraces them. The projected parabola alone therefore does not reveal direction of travel or elapsed time; the parametric equations are needed for that information.

B. Position of AV2 relative to AV1

The inverse of the first transform in Eq. (2) is

$${}^1\mathbf{T}_w = \begin{bmatrix} \mathbf{R}_z(-t) & -\mathbf{R}_z(-t)\,{}^w\mathbf{o}_1 \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}. \tag{4}$$

Therefore

$${}^1\mathbf{T}_2 = {}^1\mathbf{T}_w\,{}^w\mathbf{T}_2, \tag{5}$$

$${}^1\mathbf{o}_2(t) = \mathbf{R}_z(-t)\,({}^w\mathbf{o}_2 - {}^w\mathbf{o}_1) \tag{6}$$

$$= \begin{bmatrix} \sin t\cos t - 1 \\ -\sin^2 t \\ \cos 2t \end{bmatrix}. \tag{7}$$

The inverse rotation changes both the direction and origin of the coordinate description. Expanding the trigonometric terms makes the relative motion especially simple:

$${}^1\mathbf{o}_2(t) = \begin{bmatrix} -1 + \frac{1}{2}\sin 2t \\ -\frac{1}{2} + \frac{1}{2}\cos 2t \\ \cos 2t \end{bmatrix}. \tag{8}$$

Each component repeats after π, although the individual world-frame poses repeat after 2π. Figure 5 shows this relative period over two successive cycles. It also shows that y and z rise and fall together, with z spanning twice the excursion of y.

C. Supporting plane and intrinsic coordinates

Equation (7) gives

$$z = \cos 2t = 1 - 2\sin^2 t = 2y + 1, \tag{9}$$

so the relative curve lies in

$$\Pi: \quad 2y - z + 1 = 0. \tag{10}$$

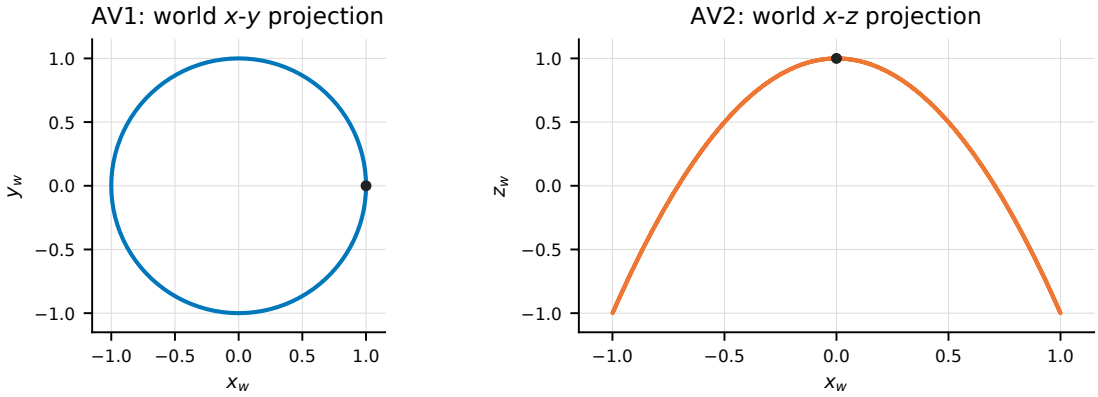


Fig. 3. Analytic world-frame projections over $0 \le t \le 2\pi$. Black markers show the starting positions at $t = 0$; the curves are calculated from Eq. (1), not extracted from RViz.

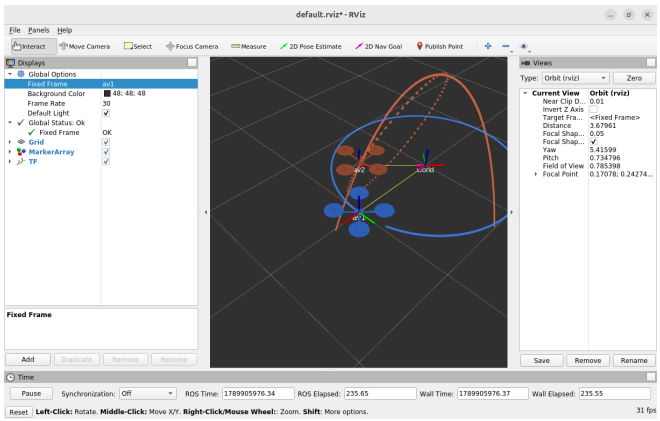


Fig. 4. RViz visualization with the fixed frame set to `av1`.

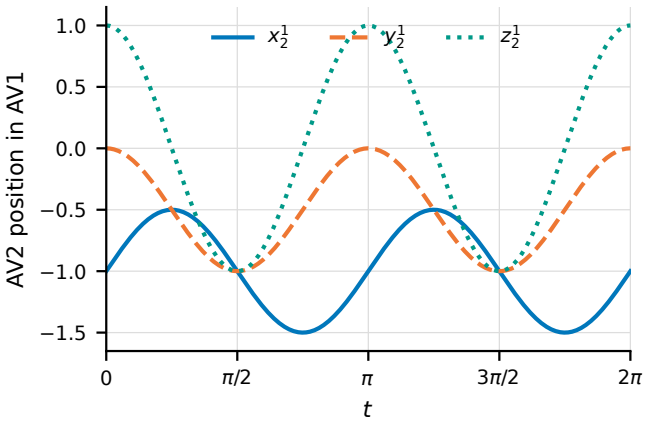


Fig. 5. Analytic components of AV2's position in AV1 from Eq. (8). The interval $0 \le t \le 2\pi$ contains two relative-motion cycles.

The plane normal is proportional to $(0, 2, -1)^\mathsf{T}$. Since the x component of this normal is zero, the plane contains lines parallel to the AV1 x axis; the linear y-z relation sets its tilt. Choose the plane origin $\mathbf{p}^1 = (-1, -1/2, 0)^\mathsf{T}$ and the orthonormal basis

$$\begin{aligned} \mathbf{e}_{x_p} &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^\mathsf{T}, \\ \mathbf{e}_{y_p} &= \frac{1}{\sqrt{5}} \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}^\mathsf{T}, \\ \mathbf{e}_{z_p} &= \frac{1}{\sqrt{5}} \begin{bmatrix} 0 & -2 & 1 \end{bmatrix}^\mathsf{T}. \end{aligned} \tag{11}$$

The centered displacement is

$${}^1\mathbf{o}_2 - \mathbf{p}^1 = \begin{bmatrix} \frac{1}{2}\sin 2t & \frac{1}{2}\cos 2t & \cos 2t \end{bmatrix}^\mathsf{T}. \tag{12}$$

Projection onto the plane basis yields

$$x_p = \frac{1}{2}\sin 2t, \qquad y_p = \frac{\sqrt{5}}{2}\cos 2t, \qquad z_p = 0. \tag{13}$$

Eliminating t gives

$$\frac{x_p^2}{(1/2)^2} + \frac{y_p^2}{(\sqrt{5}/2)^2} = 1. \tag{14}$$

Hence the relative trajectory is an ellipse centered at $\mathbf{p}^1$, with semi-axis lengths $1/2$ and $\sqrt{5}/2$. The longer semi-axis lies along $\mathbf{e}_{y_p}$, a direction tilted in the AV1 y-z plane. Figure 6 plots the ellipse in these intrinsic coordinates and projects the six saved TF positions from Table II onto the same axes. The points fall near the curve at the resolution of the saved coordinates; unlike the continuous black line, they are individual runtime samples.

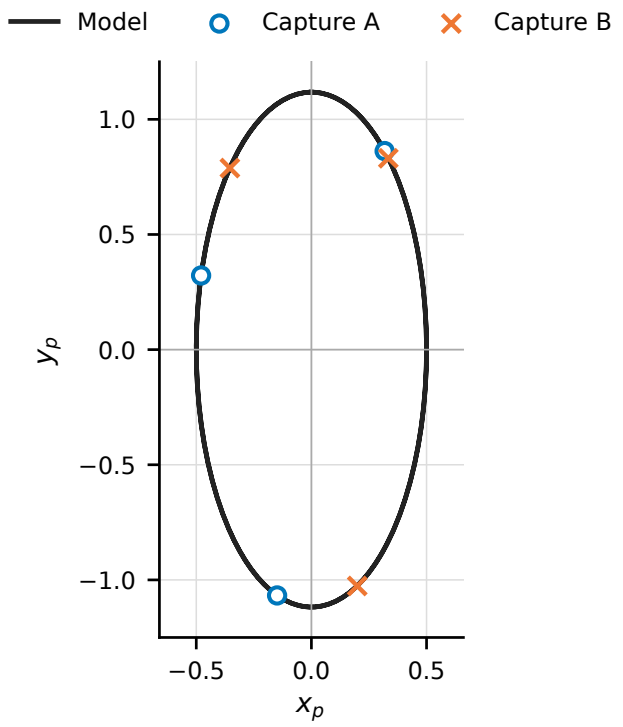


Fig. 6. Analytic relative ellipse and six TF samples in the plane coordinates of Eq. (13). Symbols distinguish the two captures in Table II.

V. Quaternion Operator Properties

For scalar-last $q = [q_1, q_2, q_3, q_4]^\mathsf{T}$, let $\Omega_1(q)$ and $\Omega_2(q)$ denote left and right quaternion multiplication.

A. Orthogonality

Direct multiplication gives, for either operator,

$$\Omega_i(q)^\mathsf{T}\Omega_i(q) = \Omega_i(q)\Omega_i(q)^\mathsf{T} = \left(q_1^2 + q_2^2 + q_3^2 + q_4^2\right)\mathbf{I}_4. \tag{15}$$

For a unit quaternion the scalar factor is one, so both matrices are orthogonal. The geometric reason is norm multiplicativity:

$$\|q \otimes x\| = \|q\|\,\|x\| = \|x\|, \tag{16}$$

and likewise for right multiplication. A linear map preserving every Euclidean norm is orthogonal.

B. Mapping to the identity quaternion

For unit q, orthogonality and the multiplication definition imply

$$\Omega_1(q)^\mathsf{T} q = q^{-1} \otimes q = \begin{bmatrix}0 & 0 & 0 & 1\end{bmatrix}^\mathsf{T}, \tag{17}$$

$$\Omega_2(q)^\mathsf{T} q = q \otimes q^{-1} = \begin{bmatrix}0 & 0 & 0 & 1\end{bmatrix}^\mathsf{T}. \tag{18}$$

Thus both transposed operators map q to the identity rotation quaternion.

C. Commutation of left and right multiplication

For arbitrary $x, y, z \in \mathbb{R}^4$, quaternion associativity gives

$$\Omega_1(x)\Omega_2(y)z = x \otimes (z \otimes y) \tag{19}$$

$$= (x \otimes z) \otimes y = \Omega_2(y)\Omega_1(x)z. \tag{20}$$

Because this holds for every z,

$$\Omega_1(x)\Omega_2(y) = \Omega_2(y)\Omega_1(x). \tag{21}$$

Furthermore, transposition corresponds to right multiplication by the conjugate: $\Omega_2(y)^\mathsf{T} = \Omega_2(y^*)$. Replacing y by y^* in the preceding argument yields

$$\Omega_1(x)\Omega_2(y)^\mathsf{T} = \Omega_2(y)^\mathsf{T}\Omega_1(x). \tag{22}$$

VI. Conclusion

The ROS implementation produced the prescribed world-frame circle and parabolic arc. Expressing AV2 in AV1 gives a shorter-period relative motion: an ellipse on $2y - z + 1 = 0$ with semi-axes $1/2$ and $\sqrt{5}/2$. Analytic plots clarify the RViz geometry, and saved TF positions sample the relative path.

References

[1] MIT SPARK Lab, "VNAV Lab 2: Exercises," 2023. [Online]. Available: https://vnav.mit.edu/labs_2023/lab2/exercises.html

[2] Open Robotics, "tf2: The second generation of the transform library." [Online]. Available: https://wiki.ros.org/tf2